Case Study: Modeling and Forecasting WTI Crude Oil Prices Using ARIMA and ARIMA-GARCH

[1] Chutimon Nunthasena
[1] Data Science, Kasetsart University, Thailand
[1] Chutimon.nun@ku.th

Abstract — This study evaluates the forecasting performance of West Texas Intermediate (WTI) crude oil prices using ARIMA and ARIMA–GARCH models. Daily observations from January 2000 to April 2024 were aggregated into monthly averages and divided into a training period (2000–2022) and a testing period (2023–2024). Stationarity was assessed using the Augmented Dickey–Fuller test; the original series was non-stationary, while the first-differenced series was stationary, supporting ARIMA with integration order $d = 1$.

Candidate ARIMA specifications were compared using the Akaike Information Criterion (AIC), where ARIMA(1,1,0) was selected as the replication baseline. To account for volatility clustering, a GARCH(1,1) model with an ARMA(1,0) mean specification (consistent with ARIMA(1,1,0) in levels) was fitted to the first-differenced series. As extensions, higher-order ARIMA(p,1,q) models were searched over $p, q \in [0, 8]$, and rolling-origin cross-validation was conducted for 3- and 6-month horizons.

Forecast accuracy was evaluated using holdout multi-step forecasting, where models were fitted on the training set and used to forecast the entire test period. Performance was measured using MSE, MAE, RMSE, and MAPE. ARIMA–GARCH yields modest improvements over the baseline ARIMA specification; however, the random-walk benchmark (RW) remains competitive and attains the lowest test error, consistent with near-random-walk dynamics in monthly WTI prices.

Index Terms— ARIMA, ARIMA–GARCH, crude oil price, time series forecasting, volatility modeling.

I. INTRODUCTION

Crude oil prices play a crucial role in the global economy, affecting inflation, exchange rates, transportation costs, and financial markets. West Texas Intermediate (WTI) is one of the most widely used benchmark crude oil prices. Due to its high volatility and sensitivity to geopolitical and macroeconomic events, accurate modeling and forecasting of WTI prices remain challenging in time series analysis.

The Autoregressive Integrated Moving Average (ARIMA) model is commonly used for forecasting economic and financial time series. However, crude oil prices often exhibit volatility clustering, where periods of large fluctuations are followed by large fluctuations and periods of small fluctuations are followed by small ones. Such conditional heteroskedasticity cannot be fully captured by ARIMA alone. To address this issue, the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model is incorporated to model time-varying variance.

This study evaluates the forecasting performance of ARIMA and ARIMA–GARCH models using monthly WTI prices. First, a replication framework is implemented using selected ARIMA specifications and evaluated through holdout multi-step forecasting. Second, an ARIMA–GARCH model is estimated to account for volatility dynamics. As an extension, higher-order ARIMA(p,1,q) models are searched over a wider lag range and assessed using both Akaike Information Criterion (AIC) and out-of-sample accuracy metrics. Rolling-origin cross-validation is additionally conducted for 3- and 6-month horizons to examine robustness.

By comparing baseline ARIMA, ARIMA–GARCH, extended high-order ARIMA, and a random-walk benchmark (RW), this study highlights the trade-off between model complexity, volatility modeling, and predictive stability in crude oil price forecasting.

II. DATA DESCRIPTION AND PREPROCESSING

A. Data Source

The dataset consists of daily West Texas Intermediate (WTI) crude oil prices obtained from the Federal Reserve Economic Data (FRED) database. The sample spans from January 2000 to April 2024 and contains 6,346 daily observations prior to aggregation.

To reduce high-frequency noise and focus on medium-term price dynamics, daily prices were aggregated into monthly averages. The final dataset consists of 292 monthly observations covering January 2000 to April 2024. Figure 1 illustrates the monthly WTI crude oil price series.

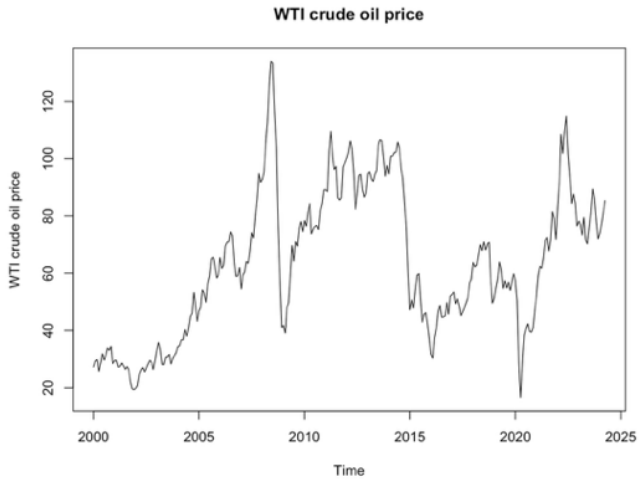

Figure 1: Monthly WTI crude oil price (2000–2024)

B. Data Transformation and Stationarity

The monthly average prices were converted into a time series object with frequency 12. The sample was divided into:

- Training set: January 2000 – December 2022 (276 observations)
- Testing set: January 2023 – April 2024 (16 observations)

This split allows for holdout multi-step forecasting evaluation.

Stationarity was assessed using the Augmented Dickey–Fuller (ADF) test on the training series. The original series produced a Dickey–Fuller statistic of −2.33 (lag order = 6, p-value = 0.44), indicating non-stationarity. After first-order differencing, the ADF statistic became −6.71 (lag order = 6, p-value < 0.01), confirming stationarity and supporting the use of ARIMA with integration order $d = 1$.

Table I. ADF

	DF	Lag	P-value
Original series	−2.33	6	0.44
First difference	−6.71	6	0.01

C. Exploratory Analysis

Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots were examined for both the original and differenced series. The original series exhibited strong persistence, while the differenced series showed short-term serial dependence.

Additionally, the squared differenced series and squared ARIMA residuals were analyzed to detect volatility clustering. The presence of autocorrelation in squared terms suggested conditional heteroskedasticity, motivating the inclusion of a GARCH component in subsequent modeling.

III. METHODOLOGY

A. ARIMA Model

To model the dynamics of monthly crude oil prices, the Autoregressive Integrated Moving Average (ARIMA) model was employed. Since the original series is non-stationary, first-order differencing was applied:

$$\Delta y_t = y_t - y_{t-1}$$

An ARIMA(p,1,q) model for the differenced series can be written as:

$$\Delta_{yt} = \phi_1 \Delta_{yt-1} + \phi_2 \Delta_{yt-2} + \cdots + \phi_p \Delta_{yt-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

where $\varepsilon_t \sim N(0, \sigma^2)$

Candidate ARIMA models were compared using the Akaike Information Criterion (AIC). In the replication stage, ARIMA(1,1,0) was selected as the baseline model.

In the extension stage, a broader search over $p, q \in [0,8]$ identified ARIMA(8,1,2) as the model with the lowest in-sample AIC.

Table II. AIC

Model	AIC
ARIMA(0,1,1)	1722.59
ARIMA(0,1,6)	1719.56
ARIMA(1,1,0)	1717.95
ARIMA(6,1,0)	1718.20
ARIMA(6,1,1)	1719.89
ARIMA(8,1,2)	1711.54

Although ARIMA(8,1,2) achieves the lowest AIC in-sample, its forecasting performance must be validated using out-of-sample evaluation.

B. GARCH Model

To capture volatility clustering, a GARCH(1,1) model was combined with the ARIMA mean equation. The error term is defined as:

$$\varepsilon_t = z_t \sqrt{h_t} \quad z_t \sim N(0,1)$$

The conditional variance follows:

$$h_t = \omega + \alpha \varepsilon_{t-1}^2 + \beta h_{t-1}$$

In this study, the GARCH(1,1) model was estimated on the first-differenced training series, with an ARMA(1,0) mean specification corresponding to ARIMA(1,1,0) in levels.

C. Model Selection and Forecast Evaluation

Model selection was based on the Akaike Information Criterion (AIC):

$$\text{AIC} = -2 \log(L) + 2k$$

Forecast accuracy was evaluated using holdout multi-step forecasting, where models were estimated on the training set (2000–2022) and used to forecast the entire test period (2023–2024).

Forecast accuracy was evaluated using:

1. Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum_{t=1}^{n} (y_t - \hat{y}_t)^2$$

2. Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_{t=1}^{n} |y_t - \hat{y}_t|$$

3. Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{MSE}$$

4. Mean Absolute Percentage Error (MAPE)

$$MAPE = \frac{100}{n} \sum_{t=1}^{n} \frac{|y_t - \hat{y}_t|}{y_t}$$

In addition, rolling-origin cross-validation was conducted for 3- and 6-month forecast horizons to assess robustness across different prediction lengths.

Table III: Shapiro-Wilk

Model	W	p-value
ARIMA(1,1,0)	0.9780	2.9e−04
ARIMA(6,1,0)	0.9800	6.6e−04

IV. RESULTS AND DISCUSSION

A. Holdout Multi-Step Forecast Accuracy

Table IV reports the forecasting performance of the competing models evaluated using holdout multi-step forecasting on the test period (January 2023 – April 2024).

Table IV. Holdout Multi-Step Forecast Accuracy

Model	MSE	MAE	RMSE	MAPE
ARIMA(1,1,0)	61.80	6.14	7.86	7.49
ARIMA(6,1,0)	31.83	4.63	5.64	6.07
ARMA(1,0)-GARCH(1,1)	31.58	4.45	5.62	5.51

The baseline ARIMA(1,1,0) exhibits the highest forecast error among the models considered. Increasing the autoregressive order to ARIMA(6,1,0) substantially improves performance, reducing RMSE from 7.86 to 5.64. This suggests that additional lag structure captures short-term dynamics more effectively.

Further improvement is observed when incorporating conditional heteroskedasticity through the GARCH(1,1) specification. The ARMA(1,0)-GARCH(1,1) model achieves the lowest error among the ARIMA-based models, with RMSE = 5.62 and MAPE = 5.51. This indicates that modeling time-varying volatility enhances predictive performance relative to mean-only specifications.

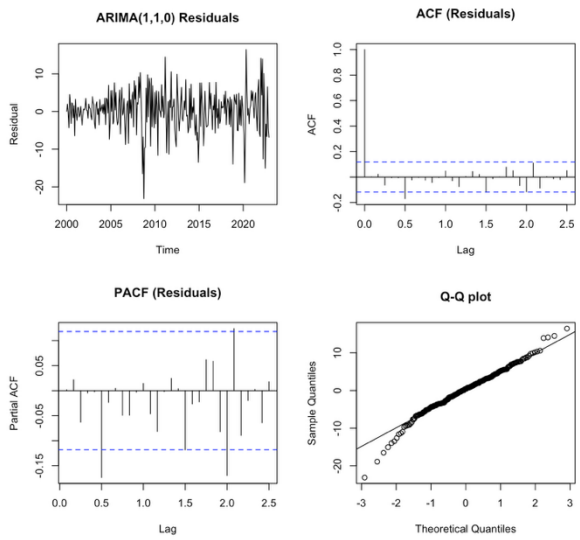

Figure 2. Residual Diagnostics for ARIMA(1,1,0)

Figure 2 presents the residual diagnostics of the ARIMA(1,1,0) model. The residual series fluctuates around zero without a clear trend. The ACF and PACF of residuals show no significant autocorrelation, indicating that the mean dynamics are adequately captured. However, the Q-Q plot reveals mild deviations from normality in the tails, suggesting the presence of heavy-tailed behavior. These findings motivate the incorporation of a GARCH component to model time-varying volatility.

B. Extended ARIMA model (AIC-best) on holdout test.

Although ARIMA(8,1,2) achieved the lowest in-sample AIC (1711.54), its holdout multi-step forecasting performance deteriorated relative to simpler specifications. On the test period, ARIMA(8,1,2) produced MSE = 47.58, MAE = 5.82, RMSE = 6.90, and MAPE = 7.70, which is worse than ARIMA(6,1,0) and ARIMA–GARCH. This gap between in-sample fit and out-of-sample accuracy suggests overfitting and reinforces the need for out-of-sample validation.

Table V. Extended ARIMA (AIC-best)

Model	MSE	MAE	RMSE	MAPE
ARIMA(8,1,2)	47.58	5.82	6.90	7.70

This result further confirms that information-criterion-based selection alone is insufficient for forecasting performance assessment.

C. Comparison with random-walk benchmark (RW)

To assess whether complex models outperform a simple baseline, a random-walk benchmark (RW) forecast was also evaluated.

Table VI. Comparison with Random-walk Benchmark (RW)

Model	MSE	MAE	RMSE	MAPE
random-walk benchmark (RW)	30.80	4.34	5.55	5.42
ARMA(1,0)-GARCH(1,1)	31.58	4.45	5.62	5.51

Interestingly, the random-walk benchmark (RW) achieves the best overall performance in the test period. The RMSE of 5.55 is slightly lower than that of the ARIMA–GARCH model. This result suggests that monthly WTI prices exhibit behavior close to a random walk, limiting the incremental gains from parametric time series modeling.

This finding underscores the importance of benchmark comparison in forecasting studies, as improvements over baseline models are often modest in financial time series.

D. Volatility Persistence

The estimated volatility persistence is:

$$\alpha_1 + \beta_1 = 0.998999$$

which is extremely close to unity. In a GARCH(1,1) framework, a persistence value near one indicates highly persistent conditional variance dynamics. This implies that shocks to volatility decay very slowly over time.

Economically, this suggests that periods of heightened uncertainty in crude oil markets tend to persist for extended durations. The strong volatility clustering observed in the squared differenced series and residuals is consistent with this near-integrated GARCH behavior.

Although incorporating volatility dynamics improves short-term predictive performance, the near-unit persistence indicates that volatility clustering is long-lasting, which may limit further gains in medium-term forecasting. This value is computed from the estimated GARCH(1,1) parameters in the fitted variance equation.

E. Rolling-Origin Cross-Validation Procedure

Rolling-origin cross-validation was implemented on the training set to evaluate forecast robustness across multiple origins. At each step, the model was re-estimated using data up to time t, and forecasts were generated for horizons of 3 and 6 months ahead.

Forecast errors were computed for each rolling origin, and the Root Mean Squared Error (RMSE) was calculated at each forecast horizon. This procedure allows performance assessment across varying forecast origins rather than relying on a single holdout split.

The results indicate that forecast errors increase substantially as the prediction horizon extends from 3 to 6 months, confirming that predictive uncertainty grows rapidly over longer horizons.

Table VII. Rolling-Origin CV

Horizon	RMSE
3 months	13.37
6 months	20.58

The rolling-origin cross-validation results indicate a substantial deterioration in forecast accuracy as the prediction horizon increases. Specifically, RMSE increases from 13.37 at the 3-month horizon to 20.58 at the 6-month horizon, reflecting rapidly growing uncertainty in medium-term forecasts.

This result suggests that predictive performance is considerably more stable in short-term horizons, while longer-term forecasts become increasingly unreliable.

Table VIII. 6-month forecasts (ARIMA)

Time	forecast	L 80	H 80	L 95	H 95
2024/05	86.73	79.75	93.71	76.05	97.41
2024/06	87.20	75.53	98.87	69.35	105.05
2024/07	87.36	71.88	102.83	63.69	111.02
2024/08	87.41	68.75	106.08	58.87	115.96
2024/09	87.43	66.01	108.86	54.66	120.20
2024/10	87.44	63.55	111.32	50.91	123.97

Table IX. GARCH forecasts

Time	Forecast	Sigma
2024/05	86.63	4.77
2024/06	87.18	4.84
2024/07	87.55	4.90
2024/08	87.87	4.96
2024/09	88.18	5.02
2024/10	88.49	5.08

F. Forecast Behavior Analysis

The forecast comparison reveals distinct differences between mean-only and variance-augmented models. The baseline ARIMA(1,1,0) produces relatively smooth forecasts under the assumption of constant variance. Incorporating a GARCH component improves short-term predictive performance by accounting for time-varying volatility.

However, the overall improvement remains modest when compared to the random-walk benchmark (RW).

This suggests that at the monthly frequency, WTI crude oil prices behave similarly to a random walk, limiting the predictive gains achievable through parametric modeling.

The cross-validation results further confirm that forecast accuracy deteriorates rapidly as the prediction horizon increases. While short-term forecasts may provide limited predictive value, medium-term forecasts remain highly uncertain.

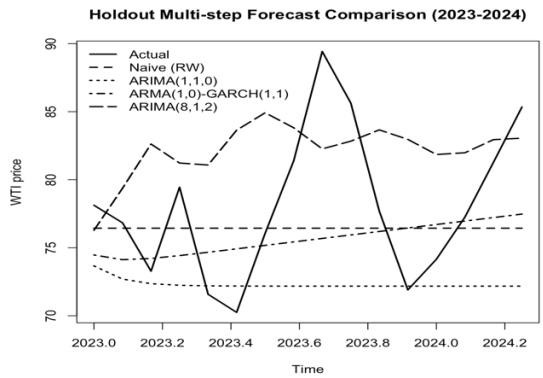

Figure 3. Holdout Multi-step Forecast Comparison (2023–2024)

G. Extended Lag Search Analysis

In the extension stage, ARIMA models were systematically searched over lag orders $p, q \in [0,8]$ while keeping the integration order fixed at one. For each candidate specification, parameters were estimated via maximum likelihood, and model selection was based on the Akaike Information Criterion (AIC).

Among all candidate models, ARIMA(8,1,2) achieved the lowest in-sample AIC (1711.54), indicating the best goodness-of-fit after penalizing model complexity. However, when evaluated using holdout multi-step forecasting, the extended model did not outperform simpler alternatives.

This result indicates that improvements in in-sample fit do not necessarily translate into improved predictive performance. The discrepancy highlights the importance of validating models using out-of-sample data rather than relying solely on information criteria.

V. CONCLUSION

This study evaluated ARIMA and ARIMA–GARCH models for forecasting monthly WTI crude oil prices from 2000 to 2024. First-order differencing successfully achieved stationarity.

While incorporating conditional heteroskedasticity improves performance relative to baseline ARIMA specifications, the random-walk benchmark (RW) achieves the strongest out-of-sample results. This finding suggests that monthly crude oil prices exhibit behavior close to a random walk, limiting the incremental gains from more complex parametric models.

Although higher-order ARIMA models achieve lower in-sample AIC values, they do not translate into improved forecasting performance, highlighting the risk of overfitting. These results emphasize the importance of model parsimony and rigorous out-of-sample validation in financial time series forecasting.

VI. LIMITATIONS AND FUTURE RESEARCH

Despite the improved performance of ARIMA-GARCH in short-term forecasting, several limitations remain. First, the model does not explicitly account for structural breaks such as the 2008 financial crisis or the COVID-19 pandemic. Second, the assumption of normally distributed innovations may underestimate extreme price movements. Third, no exogenous macroeconomic variables were incorporated into the model. Future research may explore regime-switching models, heavy-tailed distributions, or machine learning approaches to enhance predictive performance.

REFERENCES

[1] Federal Reserve Bank of St. Louis, "Crude Oil Prices: West Texas Intermediate (WTI) – Cushing, Oklahoma (DCOILWTICO)," FRED Economic Data, 2024. [Online]. Available: https://fred.stlouisfed.org/series/DCOILWTICO

[2] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed. Hoboken, NJ, USA: Wiley, 2015.

[3] C. W. J. Granger and P. Newbold, *Forecasting Economic Time Series*, 2nd ed. San Diego, CA, USA: Academic Press, 1986.

[4] R. F. Engle, "Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation," *Econometrica*, vol. 50, no. 4, pp. 987–1007, 1982.

[5] T. Bollerslev, "Generalized autoregressive conditional heteroskedasticity," *Journal of Econometrics*, vol. 31, no. 3, pp. 307–327, 1986.

[6] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne, Australia: OTexts, 2021. [Online]. Available: https://otexts.com/fpp3/

[7] R. J. Hyndman and Y. Khandakar, "Automatic time series forecasting: The forecast package for R," *Journal of Statistical Software*, vol. 27, no. 3, pp. 1–22, 2008.

[8] A. Ghalanos, "rugarch: Univariate GARCH models," R package version 1.4-9, 2023. [Online]. Available: https://CRAN.R-project.org/package=rugarch